ZCOS ARCHITECTURE FOUNDATION

Locked system architecture for the ZCOS rebuild

System: Zebulon Commander Operating System (ZCOS)

Status: Locked architecture compilation

Date: August 7, 2026

Scope: ZCOS governance, eight galaxies, seven shared galaxy domains, and rebuild boundaries

Governing rule: One ZCOS. Eight distinct galaxies. One unified Identity. Central Memory and Knowledge engines with galaxy partitions. Controlled cross-galaxy access. Seven shared domains in every galaxy.

1. Authority and Purpose

This document compiles the architecture decisions confirmed for the rebuild. It is the governing architecture wherever the prior ZebCom or ZAR specifications conflict with a locked decision here. Existing specifications remain valuable as implementation audits, migration references, and sources of reusable system logic; they do not override this architecture.

This document defines what ZCOS owns, how the eight galaxies are organized, what appears in every galaxy, and where system authority belongs. It does not certify that the current repository already implements these requirements.

Source Basis

- Zebulon Commander Operating System (ZCOS) Foundation Specification v1 (ZebCom_Spec.pdf).
- ZAR System Specification, verified August 4, 2026 (ZAR_spec_08042026.pdf).
- ZAR + User Interaction Guidelines and Behavior.
- Architecture decisions confirmed in the current ZCOS compilation session through August 7, 2026.

2. Core Architecture

ZCOS is the governing system above every galaxy. It coordinates shared intelligence, identity, memory, knowledge, projects, authorizations, and movement across the ecosystem. A galaxy is a distinct specialized system within ZCOS, not an isolated application with duplicate identity or independent foundational engines.

Locked Principles

- One intelligence, many specialized interfaces.
- Identity exists once at the ZCOS level and is referenced from every galaxy.
- Memory and Knowledge are separate authorities, even when they reuse a shared object framework.
- Memory and Knowledge each use one central ZCOS engine with eight galaxy partitions.
- Cross-galaxy access is never implicit; it is governed by ZCOS Command Desk -> Admin Access.
- Extensions are installed once for the unified Identity and become available from every galaxy.
- The shared domains have consistent purposes, while each galaxy's Desk remains specialized.
- Portal is transport, not a content or administration surface.

3. ZCOS Command Desk

The Command Desk is the system-wide control and visibility surface above the galaxies. It does not erase galaxy ownership or partition boundaries.

Command surface	Purpose
All Memory	Unified administrative view across all galaxy Memory partitions.
All Knowledge	Unified administrative view across all galaxy Knowledge partitions.
All Projects	Unified view of work created through galaxy Desks.
Admin Access	Authorization, installation, access, revocation, and audit governance.

Admin Access Owns

- Galaxy authorizations and cross-galaxy access rules.
- Read, write, and contribution authority between Memory partitions.
- Read, write, and contribution authority between Knowledge partitions.
- Authorization scope, duration, revocation, and audit records.
- Explicit promotion or sharing of Memory and Knowledge across galaxy boundaries.
- Extension installation, removal, permissions, and access.
- System administration that does not belong in an individual galaxy's Settings surface.

4. Eight Galaxies

Each galaxy combines a system identity, a console, and one specialized Desk. The seven shared domains appear in every galaxy; the Desk contents are the galaxy-specific operational center.

Galaxy	Console	Desk	Desk surfaces
ZAR	Nexys	Operate	Support; Brainstorm/Research; Tasks (Implement)
ZYNC	Canvas	Build	Coding; Design; Publish

Galaxy	Console	Desk	Desk surfaces
ZETA	Control	Integrity	Logs; Diagnostics; Monitoring
ZENO	Unite	Forum	Threads; Notes; Rooms
ZYLO	Compass	Automate	Flows/Loops; Skills; Templates
ZWAP!	Discovery	Explore	Glow; News; Journal/Blog
ZENITH	Logos	Scholar	Learning Studio; Library; Files
ZILLION	Prosper	Capital	Budgeting; Trading; Investing

5. Seven Shared Galaxy Domains

1. **Identity -** The unified user identity surface.
2. **Memory -** The galaxy-aware personal and experiential retention surface.
3. **Knowledge -** The galaxy-aware substantiated understanding surface.
4. **Apps -** The universal Extension access layer.
5. **Desk -** The galaxy's specialized working domain.
6. **Settings -** The simple control surface for account, system, appearance, and account actions.
7. **Portal -** The transport layer between the current galaxy, the ZCOS constellation, other galaxies, and ZCOS Command.

6. Identity

Identity is unified across ZCOS. Every galaxy may display the Identity domain, but no galaxy creates a duplicate user identity.

- Full name
- Preferred name
- Profile image
- Email address
- Privy wallet address

Boundary: Profile may be opened from Settings for convenience, but Identity remains the owner of profile data.

7. Memory

Memory is owned by one ZCOS Memory Engine. The existing Object Memory system evolves into its structural foundation instead of being replaced. The engine adds partitioning, lifecycle governance, authorization, retrieval controls, correction, and deletion around the object-oriented record model.

7.1 Partition Model

- Eight distinct partitions: ZAR, ZYNC, ZETA, ZENO, ZYLO, ZWAP!, ZENITH, and ZILLION.
- Each galaxy reads and writes its own partition by default.
- Every memory retains its originating galaxy and provenance.
- All Memory provides the unified ZCOS view without dissolving partition boundaries.
- Admin Access governs every cross-galaxy authorization.

7.2 User-Facing Memory

The interface remains deliberately simple. You, Topics, and Galaxies organize the same canonical records; they are not separate stores.

Area	Purpose
You	Personal experiences, decisions, people and relationships, and user-directed memories.
Topics	Memories grouped around recurring subjects, work, projects, and ongoing situations.
Galaxies	Entry to each galaxy's actual Memory partition.

The direct input Tell ZAR what to remember allows an intentional user-directed memory to enter as Confirmed.

7.3 Canonical Memory Record

- Memory content and memory type.
- Memory types: experience, decision, person/relationship, event, and user-directed memory.
- Owner user ID and originating galaxy ID.
- Source type and source ID: conversation, action, file, or system event.
- Topics and linked people, projects, or entities.
- Date the event occurred and dates the record was created or updated.
- Lifecycle state and confirmation method.
- Relationships to records it derives from, supersedes, or affects.
- Retention and deletion metadata.

7.4 Lifecycle

State	Meaning
Proposed	ZCOS inferred a candidate worth remembering.
Active	Retained under the user's approved Memory settings.

State	Meaning
Confirmed	Explicitly confirmed by the user.
Corrected	Updated while preserving correction history.
Superseded	Replaced by newer information.
Rejected	Determined not to be retained Memory.
Forgotten	Removed with downstream deletion handling.

7.5 Engine Layers and Rules

- Source ledger - where the information came from.
- Canonical memory store - the authoritative object records inside each partition.
- Retrieval indexes - topic, entity, date, relationship, semantic, and keyword indexes.
- Audit ledger - confirmation, correction, access, authorization, and deletion actions.
- Vector search is an index only; it is never the authoritative memory store.
- Corrections and deletions cascade through derived records and retrieval indexes.
- Conversations are sources and history, not automatic long-term memories.

7.6 Memory Toggle

Settings -> Memory contains Enable Memory, Memory Summary, and Manage Memory. When Memory is off, ZCOS creates no new long-term memories and does not retrieve existing memories during responses. Existing memory remains preserved unless the user deletes it. Conversation history follows separate retention controls. Re-enabling Memory restores authorized access to previously retained memory and does not silently reconstruct memories from conversations created while Memory was disabled.

8. Knowledge

Knowledge is owned by one ZCOS Knowledge Engine with eight galaxy Knowledge partitions. It mirrors Memory structurally while remaining a separate authority: Memory preserves experience; Knowledge holds what ZCOS understands and can substantiate.

8.1 Knowledge Surfaces

Surface	Purpose
Topics	Knowledge organized by subject.
Knowledge Map	Concepts, facts, claims, rules, systems, and relationships.
Sources	UGC/Uploaded or Extracted/Compiled origin, plus evidence and provenance.
Lexicon	Meanings, terminology, slang, symbols, and contextual language.
Curation	Conflicts, duplicates, confidence, currency, gaps, and open questions.

8.2 Knowledge Governance

- Eight partitions: ZAR, ZYNC, ZETA, ZENO, ZYLO, ZWAP!, ZENITH, and ZILLION.
- Knowledge retains its originating galaxy and source record.
- Each galaxy accesses its own Knowledge partition by default.
- All Knowledge provides the unified ZCOS view.
- Admin Access controls cross-galaxy Knowledge authorization.
- Origin is metadata under Sources, not a sixth Knowledge surface.

9. Shared Object Framework

Memory and Knowledge reuse a shared object framework without collapsing into one canonical authority. Object IDs, typed properties, sources, evidence, relationships, confidence, provenance, topic/entity indexing, conflict detection, and version history can be shared infrastructure.

Authority	Canonical objects
Memory	Experiences, decisions, people/relationships, events, and user-directed memories.
Knowledge	Facts, claims, concepts, systems, rules, sources, and substantiated relationships.

Objects that belong elsewhere must be routed instead of retained in an undifferentiated graph: profile data to Identity; explicit controls to Settings; inferred adaptation to Glow; tasks and work products to Desk/Projects; files to ZENITH Logos -> Scholar -> Files.

10. Apps as Extensions

Apps is the universal Extension access layer. Extensions are not galaxy-owned. The user selects Add Extension once under the unified Identity; the installed Extension then becomes available from every galaxy's Apps domain.

Extension	Function
PoeTrees Music	Stream
Zupreme Imports	Shop
Property Pulse	Housing
Self-Help Simplified	eBooks
EmojiLingo	New Language
Fantasma Firewall / FanFI	Security Firewall
Heritage Haven	Ancestry Programs
Forensic Vision	Deepfake, edited, photoshopped, masked, and generated-media detection

Extension	Function
Good Neighbor	Landlord Review Service

Boundary: Apps contains ZCOS Extensions. Settings -> Integrations connects external services. These are separate systems.

11. Desk and Projects

Desk is each galaxy's specialized working domain. Desk contents are not shared across galaxies. Work created through a Desk becomes a Project associated with that galaxy, while ZCOS Command Desk -> All Projects provides the unified project view.

- Every galaxy has exactly one Desk.
- Each Desk contains the three confirmed surfaces listed in the Eight Galaxies table.
- Existing repository workspaces and feature pages are migration sources; they do not redefine the locked Desk architecture.
- Cross-galaxy project authority is governed through Admin Access.

12. Settings

Settings uses a simple grouped-card layout with plain labels, large touch targets, and dedicated subpages. The visible interface remains simple even when the governed system is complex.

Group	Settings
Account	Profile; Billing; Notifications; Time & Focus; Privacy; Shared Links
System	Capabilities; Integrations; Device Permissions; Voice; Memory; Interaction Style; Haptic Feedback
Appearance	Light; Dark; System
Account Actions	Log out; Delete account

12.1 Setting Ownership

- Profile is a shortcut to Identity; Identity owns the data.
- Integrations manages external services; backend connectors technically enable them.
- Device Permissions controls access to the user's device.
- Admin Access controls galaxies, Extensions, cross-galaxy authority, and system authorization.
- Memory contains Enable Memory, Memory Summary, and Manage Memory.
- Manage Memory opens You, Topics, and Galaxies.
- Interaction Style contains tone, response length, explanation depth, pace, and preferred response formats.

12.2 Confirmed Subpage Contents

Subpage	Contents
Notifications	Completed work; responses; approval requests; system alerts; product updates
Time & Focus	Break reminders; quiet hours; active days; notification schedule
Privacy	Data-use controls; conversation retention; archived conversations; export data; deletion controls; privacy policy
Shared Links	View active links; disable individual links; disable all links
Capabilities	Web research; code execution; file creation/editing; image generation; external actions; future automatic model routing
Integrations	Discover; add; connected services; connection status; permissions; disconnect
Device Permissions	Location; camera; microphone; photos; files; contacts; calendar; reminders; notifications; motion and fitness
Voice	ZAR voice; spoken language; speech speed; hands-free; push to talk

13. Portal

Portal is the transport layer used to leave the current galaxy and enter the ZCOS constellation, another available galaxy, or ZCOS Command.

- Opens the constellation view.
- Identifies the current galaxy.
- Allows entry into another available galaxy or ZCOS Command.
- Preserves the unified Identity during movement.
- Contains no content, Settings, Memory, Knowledge, or administrative controls.

14. Domain Boundaries

Domain	Owns
Identity	Who the user is and the canonical profile record.
Memory	What ZCOS preserves from experience and intentionally recalls.
Knowledge	What ZCOS understands and can substantiate.
Files	Original stored artifacts under ZENITH Logos -> Scholar.
Library	Organization and access to stored artifacts under Scholar.
Glow	Adaptation and learning about the individual under ZWAP! Discovery -> Explore.
History	What occurred within the system; not automatically long-term Memory.
Apps	Installed ZCOS Extensions.

Domain	Owns
Integrations	Connected external services.
Portal	Movement between system locations.

15. Rebuild and Migration Rules

- Preserve reusable object, provenance, graph, ingestion, curation, lexicon, retrieval, and audit components where they fit the locked owners.
- Do not keep repairing disconnected ZAR Memory or Knowledge authorities as independent long-term systems.
- Do not retain one undifferentiated Object Memory graph as the authority for Identity, Memory, Knowledge, Glow, tasks, projects, files, and Settings.
- Migrate by classification and ownership: route each existing object to its correct ZCOS domain and galaxy partition.
- Do not invent fallback owners. User-owned records require the authenticated owner identity.
- Do not treat local JSON, legacy archives, exported conversations, or vector indexes as canonical production truth.
- Do not alter the established interface design while rebuilding underlying ownership and wiring unless separately approved.

16. Required Specification Order

Compilation is complete at the architecture level. Implementation proceeds only after the dependent specifications are written against this foundation.

1. ZCOS Memory Engine Specification.
2. ZCOS Knowledge Engine Specification.
3. Rewritten ZCOS System Specification.
4. Rewritten ZAR System Specification.
5. Remaining galaxy specifications and migration plans.
6. Repository rebuild and verified implementation.

Final architecture statement: ZCOS governs one unified intelligence ecosystem. Identity is singular. Memory and Knowledge are centrally governed and partitioned by galaxy. Extensions install once and appear everywhere. Each galaxy has the same seven domains, one specialized Desk, and controlled movement through Portal.